EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 03

Practice problems in topics covered till: Lecture 09

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

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1. Using a programming language of your choice, implement the Forward Euler, Backward Euler, RK2, RK4, and Trapezoidal methods to solve an ODE of the form $\frac{dy}{dx} = f(x, y)$. For the explicit methods (Forward Euler, RK2, RK4), the program must be designed to be generic, meaning that it should take the function $f(x, y)$, the initial condition $y = y_0$, and the step size h as inputs. Specifically, the method should be implemented as a callable function where the user can provide the function $f(x, y)$, the initial condition $y = y_0$, and the step size h to solve the ODE.

   The code should be organized modularly, such that the different methods (Forward Euler, Backward Euler, RK2, RK4, and Trapezoidal) are implemented as independent functions that can be called individually.

   As an advanced programming exercise, create a module (or a class) that implements explicit methods (Forward Euler, RK2, RK4). This module should allow users to test these methods by solving different ODEs, including those provided in this course. The module should be designed so that it can be easily imported into other scripts, enabling users to test different ODEs and compare the results from various methods.

   Additionally, the program must generate an image showing how the solutions evolve over time.

2. Using the code you have developed, analyze the solution to the ODE $\frac{dy}{dx} = 1000\,(3 - 2e^{-x} - y)$ with the initial condition $y(0) = 0$ using all the numerical methods implemented (Forward Euler, Backward Euler, RK2, RK4, and Trapezoidal). Choose the step size to be 0.01. After implementing the methods, examine and compare their behavior.

3. Using the code you have developed, analyze the solution to the ODE $\frac{dy}{dx} = 1000\,(1 - e^{-x} - y)$ with the initial condition $y(0) = 0$ using all the numerical methods implemented (Forward Euler, Backward Euler, RK2, RK4, and Trapezoidal). Choose the step size to be 0.01. After implementing the methods, examine and compare their behavior.

4. A tank contains 1000 litres of brine with 100 kg of dissolved salt. Fresh water enters the tank at a rate of 10 litres per minute, and the well-mixed solution leaves the tank at the same rate. The differential equation governing the amount of salt $S(t)$ at a time t is given by $\frac{dS}{dt} = -\frac{S}{100}$.

   - Solve the equation numerically for S(t) over a period of 5 minutes using the Forward Euler Method with a fixed step size of 0.5.

- Solve the equation numerically for S(t) over a period of 5 minutes using the Runge-Kutta 4 Method with a fixed step size of 0.5.
- Compute the exact analytical solution.
- Calculate the local error at 5 minutes for both the methods.

5. Prove/disprove the following statement: For the differential equation $\frac{dy}{dx} = f(y)$, every line $y = c$ (where c is a constant) is an isocline. Furthermore, do all isoclines differ by a constant factor?.

6. Prove that the Forward Euler method is numerically stable for solving the initial value problem $\frac{dy}{dx} = \lambda y$ with $y(x_0) = y_0$, when the step size $h \leq \frac{2}{|\lambda|}$.

7. Prove that the Runge-Kutta 4th order method (RK4) is numerically stable for solving the initial value problem $\frac{dy}{dx} = \lambda y$ with $y(x_0) = y_0$, when the step size $h \leq \frac{4}{|\lambda|}$.

8. The Adams-Bashforth 2-step method is an explicit multistep method for solving initial value problems of the form. This method uses two previous points to calculate the next value, making it a two-stage method. The update equation for the Adams-Bashforth 2-step method is

$$y_{n+1} = y_n + \frac{h}{2}\left[3f(x_n, y_n) - f(x_{n-1}, y_{n-1})\right] \tag{1}$$

   Derive the condition that determines whether the numerical method is stable for a given step size.

9. Derive the general update equations for the Backward Euler method and Trapezoidal method for solving the IVP: $\frac{dy}{dx} = \exp(0.1x) - 10y$

10. Consider the non-linear equation: $x^3 - 5x + 1 = 0$. Use Newton Raphson method to find the root of this equation using an initial guess of $x_0 = 2$. Solve over 3 iterations.

Please Note: Some of the problems presented in this document have been adapted from textbooks referenced for this course. These problems are included to help students practice and understand key concepts. The material is used for educational purposes only. We have made every effort to provide proper citations, but if any are inadvertently omitted, it was not intentional.